# DEPARTEMENT STATISTIEK EN AKTUARIËLE WETENSKAP
## *DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE*

# STATISTIEK EN DATAWETENSKAP 188
## *STATISTICS AND DATA SCIENCE 188*

# TUTORIAAL 7 / *TUTORIAL* 7

Die meeste probleme is aangepas uit BASIC Business Statistics – Berenson, Levine & Krebiel, 10th ed, Prentice Hall. / *Most problems are adapted from BASIC Business Statistics – Berenson, Levine & Krebiel, 10th ed, Prentice Hall.*

1. Gegee 'n normaalverdeling met $\mu = 100$ en $\sigma = 10$. Indien 'n steekproef van $n = 25$ selekteer word, wat is die waarskynlikheid dat $\bar{X}$ *:* / *Given a normal distribution with* $\mu = 100$ *and* $\sigma = 10$ *and, if you select a sample of* $n = 25$*, what is the probability that* $\bar{X}$ *is*:
   (a) Kleiner as 95 is? / *Less than* 95*?*
   (b) Tussen 95 en 97.5 is? / *Between* 95 *and* 97.5*?*
   (c) Bo 102.2 is? / *Above* 102.2*?*
   (d) Daar is 'n 65% kans dat $\bar{X}$ bo watter waarde is? / *There is* 65*% chance that* $\bar{X}$ *is above which value?*

2. Die punte behaal in 'n statistiek toets is normaal verdeel met 'n gemiddelde van 78 en 'n variansie van 36. / *The marks obtained in a statistics test are normally distributed with a mean of* 78 *and a variance of* 36*.*
   (a) Watter proporsie van die klas het 'n punt van minder as 75 behaal? / *What proportion of the class obtained a mark below* 75*?*
   (b) Wat is die waarskynlikheid dat 'n klas van 50 studente 'n gemiddelde toetspunt van minder as 75 sal behaal? / *What is the probability that a class of* 50 *students will obtain an average test mark below* 75*?*

3. Veronderstel 'n sentrale ontvang oproepe tussen 10:00 en 11:00 teen 'n gemiddelde van 400 oproepe per uur. Aanvaar die tye tussen oproepe is eksponensiaal verdeel. / *A switchboard receives calls between 10:00 and 11:00 at an average rate of* 400 *calls per hour. Assume the times between calls to follow an exponential distribution.*
   (a) Wat is die waarskynlikheid dat die tyd tussen die volgende twee oproepe meer as 12 sekondes sal wees? / *What is the probability that the time between the next two calls will exceed* 12 *seconds?*
   (b) Gestel ons neem die volgende 30 tye tussen opeenvolgende oproepe waar. Wat is die waarskynlikheid dat die gemiddeld van hierdie 30 tye meer as 12 sekondes sal wees? / *Suppose we observe the next* 30 *times between consecutive calls. What is the probability that the mean of these* 30 *times will exceed* 12 *seconds?*


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4. Die volgende data verteenwoordig die aantal dae per jaar wat 'n populasie van ses werknemers in 'n klein maatskappy afwesig is: / *The following data represent the number of days absent per year in a population of six employees of a small company*:

1, 3, 6, 7, 9, 10.

(a) Neem aan dat u 'n steekproef sonder terugplasing trek, selekteer alle moontlike steekproewe van $n = 2$ en konstrueer die steekproefverdeling van die gemiddelde. Bereken die gemiddelde van al die steekproefgemiddeldes en bereken ook die populasiegemiddelde. Is hulle gelyk? Wat word hierdie eienskap genoem? / *Assuming that you sample without replacement, select all possible samples of* $n = 2$ *and construct the sampling distribution of the mean. Compute the mean of all the sample means and also compute the population mean. Are they equal? What is this property called?*

(b) Herhaal (a) vir alle moontlike steekproewe van $n = 3$. / *Repeat (a) for all possible samples of* $n = 3$.

(c) Vergelyk die vorms van die steekproefverdelings van die gemiddelde in (a) en (b). Watter steekproefverdeling het minder variasie? Waarom? / *Compare the shape of the sampling distribution of the mean in (a) and (b). Which sampling distribution has less variability? Why?*

Indien u bostaande eksperimente met steekproefneming met terugplasing sou herhaal, sou u soortgelyke resultate kry. / *If you perform the above experiments with sampling with replacement you would have obtained similar results.*

5. Let op dat $\sigma_p$ afhanklik is van die populasieproporsie $\pi$. / *Note that* $\sigma_p$ *is dependent on the population proportion,* $\pi$.

In 'n ewekansige steekproef van 64 persone word 48 as "suksesvol" geklassifieer. Indien die populasieproporsie 0.70 is, / *In a random sample of* 64 *people,* 48 *are classified as "successful". If the population proportion is* 0.70,

(a) bepaal die steekproefproporsie $p$ van "suksesvolle" persone. / *determine the sample proportion* $p$ *of "successful" people.*

(b) bepaal die standaardfout van die proporsie. / *determine the standard error of the proportion*.

6. Die volgende data verteenwoordig die antwoorde (Y vir ja en N vir nee) van 'n steekproef van 40 kollege studente op die vraag: "Besit u tans enige aandele?" / *The following data represent the responses (Y for yes and N for no) from a sample of* 40 *college students to the question, "Do you currently own shares in any stock?"*

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *N* | *N* | *Y* | *N* | *N* | *Y* | *N* | *Y* | *N* | *Y* | *N* | *N* | *Y* | *N* | *Y* | *Y* | *N* | *N* | *N* | *Y* |
| *N* | *Y* | *N* | *N* | *N* | *N* | *Y* | *N* | *N* | *Y* | *Y* | *N* | *N* | *N* | *Y* | *N* | *N* | *Y* | *N* | *N* |

(a) Bepaal die steekproefproporsie $p$ van kollege studente wat aandele besit. / *Determine the sample proportion* $p$ *of college students who own shares*.

(b) Beraam die standaardfout van die steekproefproporsie. / *Estimate the standard error of the sample proportion*.


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7. ’n Politieke verkiesingsopnemer doen ’n analise van steekproef resultate om voorspellings te maak oor ’n verkiesing. Veronderstel daar is twee kandidate. Indien ’n spesifieke kandidaat ten minste 55% van die stemme in die steekproef kry, word voorspel dat die kandidaat sal wen. As ’n steekproef van 100 stemgeregtigdes gekies word, wat is die waarskynlikheid dat ’n kandidaat as wenner voorspel sal word indien / *A political analyst is conducting a poll in order to make predictions on election night. Assuming a two-candidate election, if a specific candidate receives at least* 55*% of the vote in the sample, then that candidate will be forecast as the winner of the election. If you select a sample of* 100 *voters, what is the probability that a candidate will be forecast as the winner when*
   (a) die werklike persentasie wat vir haar stem 50.1% is? / *the true percentage of her vote is* 50.1*%?*
   (b) die werklike persentasie wat vir haar stem 60% is? / *the true percentage of her vote is* 60*%?*
   (c) die werklike persentasie wat vir haar stem 49% is (en sy eintlik die verkiesing verloor)? / *the true percentage of her vote is* 49*% (and she will actually lose the election)?*
   (d) As die steekproefgrootte vermeerder word na 400, wat is u antwoorde op (a)? Bespreek u antwoord. / *If the sample size is increased to* 400*, what are your answers to (a)? Discuss your answer*.

8. Volgens die Nasionale Restaurant Vereniging het 20% van fynproewer restaurante ’n beleid ingestel wat die gebruik van selfone beperk (*“Business Bulletin”, The Wall Street Journal, June 1, 2000, A1*). Indien ’n ewekansige steekproef van 100 fynproewer restaurante geselekteer word, / *According to the National Restaurant Association,* 20*% of fine-dining restaurants have instituted a policy restricting the use of cell phones (“Business Bulletin”, The Wall Street Journal, June 1, 2000, A1). If you select a random sample of* 100 *fine-dining restaurants,*
   (a) wat is die waarskynlikheid dat in die steekproef tussen 15% en 25% van die restaurante ’n gevestigde beleid het wat selfoongebruik beperk? / *what is the probability that the sample has between* 15*% and* 25*% of the restaurants that have established a policy restricting cell phone use?*
   (b) is die waarskynlikheid 90% dat die steekproefpersentasie binne watter simmetriese grense vanaf die populasiepersentasie lê? / *the probability is* 90*% that the sample percentage will be contained within what symmetrical limits of the population percentage*?
   (c) is die waarskynlikheid 95% dat die steekproef persentasie tussen watter simmetriese grense bevat sal word. / *the probability is* 95*% that the sample percentage will be contained within what symmetrical limits of the population percentage?*

9. DiGiorno’s bevrore pizza het van die mees kreatiewe advertensies op televisie. ’n Koerant se advertensiekolom beweer dat 20% van alle kykers “baie” van die advertensies hou. Veronderstel dat die advertensies vir ’n steekproef van 400 televisiekykers gewys word. Wat is die waarskynlikheid dat die steekproef tussen: / *DiGiorno’s frozen pizza has some of the most creative advertisements on television. A paper’s advertisement column claims that* 20*% of viewers like the ads “a lot”. Suppose that a sample of* 400 *television viewers is shown the advertisements. What is the probability that the sample will have between*:
   (a) 18% en 22% persone wat “baie” van die advertensies hou bevat? / 18*% and* 22*% who like the ads “a lot”?*
   (b) 16% en 24% persone wat “baie” van die advertensies hou bevat? / 16*% and* 24*% who like the ads “a lot”?*
   (c) 14% en 26% wat “baie” van die advertensies hou bevat? / 14*% and* 26*% who like the ads “a lot”?*


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(d) 12% en 28% wat "baie" van die advertensies hou bevat? / 12*% and* 28*% who like the ads "a lot"?*

10. Waarom is die steekproefgemiddelde 'n onsydige beramer van die populasiegemiddelde? / *Why is the sample mean an unbiased estimator of the population mean?*

11. Waarom daal die standaardfout van die gemiddelde indien die steekproefgrootte $n$ styg? / *Why does the standard error of the mean decrease as the sample size n increases?*

12. Onafhanklike ewekansige steekproewe van tien waarnemings elk word geneem uit twee normaal populasies. Die parameters vir hierdie populasies is: */ Independent random samples of ten observations each are drawn from normal populations. The parameters of these populations are:*

Populasie 1 */ Population 1:* $\mu = 280$, $\sigma = 25$.
Populasie 2 */ Population 2:* $\mu = 270$, $\sigma = 30$.

(a) Bereken die waarskynlikheid dat die gemiddelde van steekproef 1 meer is as die gemiddelde van steekproef 2 met 25. */ Calculate the probability that the mean of sample 1 is greater than the mean of sample 2 by more than* 25*.*
(b) Herhaal (a) met 'n steekproefgrootte van 50. */ Repeat (a) for a sample size of* 50*.*
(c) Herhaal (a) met 'n steekproefgrootte van 100. */ Repeat (a) for a sample size of* 100*.*

13. 'n Statistiek dosent weet dat die punte vir sy kursus normaal verdeel is. Hy bereken dat sy oggendklasse 'n gemiddelde van 53% met 'n standaardafwyking van 10% vir hulle finale eksamen behaal het. Sy middagklasse het 'n gemiddelde van 58% met 'n standaardafwyking van 8% vir hulle finale eksamen behaal. Wat is die waarskynlikheid dat die gemiddelde punt van nege ewekansig geselekteerde studente uit die oggendklasse hoër is as die gemiddelde punt van nege ewekansig geselekteerde studente uit die middagklasse? */ A statistics lecturer knows that the marks in his course are normally distributed. He has calculated that his morning classes average* 53*% with a standard deviation of* 10*% on their final exams. His afternoon classes average* 58*% with a standard deviation of* 8*%. What is the probability that the mean mark of nine randomly selected students from a morning class is greater than the average mark of nine randomly selected students from an afternoon class?*

14. Die deursnee van Ping-Pong-balle wat in 'n groot fabriek vervaardig word, is ongeveer normaal verdeel, met 'n gemiddelde van 1.30 duim en 'n standaardafwyking van 0.04 duim. As 'n ewekansige steekproef van 16 Ping-Pong balle gekies word uit 'n populasie van 200 sonder terugplasing, wat is die waarskynlikheid dat die steekproefgemiddelde tussen 1.31 en 1.33 duim is? / *The diameter of Ping-Pong balls manufactured at a large factory is approximately normally distributed, with a mean of 1.30 inches and a standard deviation of 0.04 inches. If a random sample of 16 Ping-Pong balls is selected from a population of 200 Ping-Pong balls without replacement, what is the probability that the sample mean is between 1.31 and 1.33 inches?*


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